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Development and validation of a liquid chromatographic tandem mass spectrometric method (LC–MS/MS) for the determination of bisphenols and parabens in menstrual products after solid-phase extraction

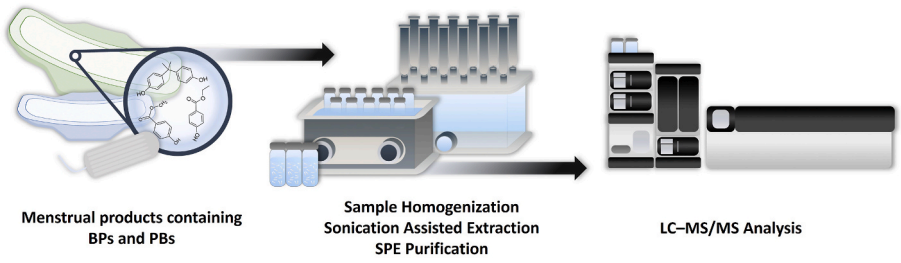
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HIGHLIGHTS

- 6 bisphenols and 5 parabens quantified in menstrual products by SPE-LC-MS/MS.
- Method showed high recoveries (84–112%), $R^2 > 0.99$, LOD/LOQ 0.2–1 ng/mL, RSD≤20%.
- BPA detected in all samples, ranging from LOQ to 68 ng/g.
- Parabens detected in pads, higher in panty liners, LOQ to 3635 ng/g.
- ComplexMoGAPI, RAPI and BAGI evaluated method greenness, robustness, applicability.

GRAPHICAL ABSTRACT



ARTICLE INFO

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SPE
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ABSTRACT

Background: Currently, thousands of personal care and hygiene products are widely produced and distributed worldwide. Even though in recent years legislation concerning these compounds has become stricter than ever before, limited information is provided on product labels.

Results: In this research, a novel method was developed and validated for the determination of parabens and bisphenols in menstrual products, including menstrual pads, panty liners, and tampons. Sample preparation was carried out using solid-phase extraction (SPE) prior to analysis by liquid chromatography coupled with tandem mass spectrometry (LC–MS/MS) equipped with an electrospray ionization (ESI) source. The optimized method showed high recoveries ranging from 84.0 to 112.0%, with relative standard deviation (RSDs) not exceeding 20.0% at the limit of quantification (LOQ) and 15.0% at the remaining concentration levels. The method was also applied to real samples, in which bisphenol A was the most prevalent compound detected, while parabens were also determined in several analysed samples.

Significance and novelty: This work introduces a rapid LC method for the simultaneous extraction, purification, and quantification of bisphenols and parabens in menstrual products, with reduced solvent consumption and analysis time. The applicability, analytical performance, and environmental sustainability of the method were evaluated using established assessment tools. The results demonstrate that the proposed method aligns with the principles of green chemistry.

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1. Introduction

The plastics and polymer industry widely employs bisphenols (BPs) monomers as key starting materials in the production of numerous polymeric products, including packaging materials. Among these compounds, BPA has been extensively studied and reported in a wide range of environmental, food and consumer product matrices [1–3]. BPA has primarily been used as a monomer in the production of polycarbonate plastics and epoxy resins since the 1950s [4]. BPs are also used as additives, acting as plasticizers, hardeners or antioxidants in certain polymer formulations [5–7]. Due to its physicochemical properties, BPs can migrate from packaging materials into the contained products or be introduced through contamination during manufacturing processes. The adverse effects of BPA, particularly its classification as a potential carcinogen and endocrine disruptor, have led to its gradual phase-out due to stricter legislation [8]. As a result, manufacturers have increasingly replaced BPA with structural analogues such as bisphenol S (BPS) among other bisphenol derivatives. Nevertheless, these compounds have also been studied and have shown similarities in endocrine-disrupting activity [3,9,10]. Despite ongoing regulations limiting BPs exposure, these compounds continue to be detected in a variety of hygiene, personal care products and environmental samples.

Parabens (PBs), on the other hand, are intentionally added to many personal care products for stabilization and preservation of active compounds, and may also function as antimicrobial and emulsifying agents [11–13]. The concentration levels of PBs are regulated by legislative frameworks, as numerous studies have demonstrated their potential link to endocrine disruption and other adverse health effects [14–16]. The most commonly detected PBs in personal care products are Methyl paraben (MeP), Ethyl paraben (EtP), Propyl paraben (PrP), Butyl paraben (BuP) and Benzyl Paraben (BenP) which are typically present in varying concentrations depending on the product formulation [17–19].

Most hygiene products contain plastic components, coatings, or adhesives that can introduce BPs residues. Menstrual pads mainly consist of a soft top layer that comes into contact with the skin. This layer is typically made from nonwoven polypropylene (PP) or polyethylene (PE), and emollients are usually added to enhance comfort [20]. It is designed with specially engineered perforations or patterns that quickly channel menstrual fluid away from the body. Beneath this layer are multiple absorbent layers that capture and retain the liquid, often containing superabsorbent polymers to improve absorption capacity. The bottom layer is made of a waterproof, hydrophobic plastic material that prevents leakage and helps keep clothing dry [21]. These synthetic materials, also found in other menstrual products, may contain and potentially release harmful substances through skin contact, including volatile organic compounds (VOCs), phthalates, fragrances and heavy metals, as well as BPs and PBs [17,20,22–24], which are also investigated in this study.

Extensive literature is available on the isolation and quantification of BPs and PBs, mainly in biological matrices such as urine, [25–27] blood [14,28,29], and other human fluids including breast milk [30,31], as well as in environmental samples such as wastewater and surface waters [32]. PBs and BPs have also been widely analysed in personal care products, including creams, shampoos, and similar formulations [33–36], however, less information is available regarding their presence in menstrual products [17,24,37]. In addition, most published methods typically rely on the use of relatively large volumes of organic solvents such as dichloromethane:hexane (1:4, v/v; ~10 mL total) or methyl tert-butyl ether (~8 mL total), or even prolonged extraction procedures using up to 40 mL of methanol [17,37,38]. These approaches require laborious procedures and are often coupled with multiple shaking, ultrasonication, and centrifugation steps, resulting in extended sample preparation times ranging from several hours to overnight. Moreover, many methods are primarily focused on extraction efficiency, with limited focus on sample clean-up and overall method optimization. To the best of our knowledge, this is among the few studies reporting the simultaneous determination of multiple bisphenols and parabens in menstrual products using a miniaturized extraction protocol combined with solid phase extraction (SPE) clean-up and LC–MS/MS analysis. In contrast to previously reported methodologies, the proposed approach substantially reduces solvent consumption and sample preparation time while maintaining satisfactory analytical performance in terms of sensitivity, recovery, precision, and matrix-effect control. Furthermore, beyond analytical validation, the method was systematically evaluated using contemporary green and practical analytical assessment tools, providing a comprehensive evaluation of its environmental sustainability, analytical quality, and applicability for routine monitoring of endocrine-disrupting chemicals in menstrual products.

In this study, a rapid, simple, and easily applicable analytical method is proposed for the simultaneous extraction, purification, and quantification of BPs and PBs in menstrual pads, panty liners, and tampons using LC, aiming to address this gap in the literature. The method employs a relatively greener solvent for extraction, MeOH, compared to chlorinated and petroleum-based solvents used in previous works, with a reduced extraction volume of only 2 mL and 15 min ultrasonication, combined with an SPE sample clean-up step. Both the extraction and purification steps were optimized for the determination of the two classes of endocrine disrupting chemicals, with the objective not only to achieve reliable quantification but to minimize organic solvent usage and type as well as overall analysis time. The developed method was further applied to real samples to determine the concentrations of these compounds, addressing the fact that product labelling often does not provide information regarding their presence or levels. The novelty of the present work lies in the development of a solvent-minimized SPE–LC–MS/MS methodology for the simultaneous determination of bisphenols and parabens in menstrual products, a matrix that remains insufficiently investigated despite its direct and prolonged contact with the human body. The proposed method combines miniaturized sample preparation, effective SPE clean-up, comprehensive validation, and green analytical chemistry assessment, offering a practical alternative to conventional extraction procedures that rely on larger solvent volumes and labor-intensive workflows. In addition, its successful application to a variety of commercially available menstrual products provides new occurrence data for endocrine-disrupting compounds.

2. Materials and methods

2.1. Materials and reagents

Analytical standards of BPs (BPS, BPF, BPE, BPA, BPB and BPC), PBs (MeP, EtP, PrP, BenP, BuP) and diethylstilbestrol (DES) were provided from the International Forensic Research Institute, Department of Chemistry and Biochemistry, Florida International University (Miami, FL, USA), and were of the highest purity (>99%). LC-MS grade acetonitrile (ACN), methanol (MeOH) and water (H_2O) were purchased from VWR Chemicals (Leuven, Belgium), while acetone was purchased from Chem-Lab (Zedelgem, Belgium). Formic acid (CH_2O_2, 98%) was purchased from TCI (Portland, USA) while ammonium hydroxide (NH_4OH) was obtained from Sigma-Aldrich (St. Louis, MO, USA). For SPE extraction and purification, five different SPE cartridges were used: OASIS HLB 3 cc, 60 mg (Waters Corporation, Milford, Massachusetts, USA); Bond Elut SAX, SCX, and FL 6 cc, 500 mg (Agilent Technologies, USA); Discovery DSC-18 3 cc, 500 mg (Supelco, Sigma-Aldrich/Merck KGaA, Darmstadt, Germany); LICHROLUT RP-18 3 cc, 200 mg (40-63 μm), (Merck KGaA, Darmstadt, Germany); and a 12-port Visiprep 57044 SPE device (Supelco, Sigma-Aldrich/Merck KGaA, Darmstadt, Germany).

2.2. Standards & sample preparation

Individual stock solutions of each analyte were prepared in MeOH MS grade at a concentration level of 100 ng/μL and stored at −20 °C

until use. A working standard mixture containing all the analytes was prepared daily by appropriate dilution of the stock solutions to yield concentrations within the validated calibration range of 1–100 ng/mL.

All real samples used for validation (blank samples) and analysis were purchased from major chain supermarkets in Thessaloniki, Greece and consisted of commonly commercially available brands. Products were transported to the laboratory in their original sealed packaging to prevent contamination and were opened only immediately prior to analysis. In addition to menstrual pads, products with similar material characteristics including sterile sealed gauze, cotton, and standard toilet paper were analysed to further evaluate the applicability of the method to comparable matrices. These materials were used as procedural blanks for BPs and PBs to demonstrate method selectivity. All products were analysed at least in triplicate ($n \geq 3$). Prior to extraction, each sample was homogenized using material obtained from one or multiple units of the same batch. As BPs and PBs are prevalent compounds, all tools (tweezers, scissors, gloves, etc.) used for maceration of the samples were thoroughly cleaned with methanol to remove any traces of analytes.

2.3. Extraction and SPE procedure

A non-scented menstrual pad was selected for the optimization procedure, as it showed no detectable levels of target analytes except for BPA, which was below the LOQ. For extraction, only the portion of the pad that comes into contact with the skin and the adsorbent layer was assessed, while the remaining layers, which have negligible skin interaction, were excluded. This section was cut into small pieces and homogenized to ensure uniformity. After that, 0.1 g of the sample was accurately weighed and transferred into a small glass vial, and was spiked with varying concentrations of mixed standards to obtain final concentrations ranging from 1 to 100 ng/mL. According to the initial conditions, 2 mL of MeOH MS was added to the sample and extracted for 15 min using an ultrasonic bath. Samples were then vortexed for 1 min, and the entire pad pieces were transferred into a sterile syringe with a filter attached and squeezed to collect the extracted sample. Due to the pad's composition, which includes cotton and other adsorptive materials, conventional solvent collection methods, such as centrifugation, proved to be insufficient. The resulting methanolic extract was then diluted with water to minimize the organic solvent percentage prior to SPE. Subsequently, SPE cartridges were first conditioned according to the manufacturer's instructions and samples were then loaded and eluted with an appropriate organic solvent. Specifically, for conditioning 2 mL of MeOH followed by 2 mL of H_2O were used with a flow rate of 1 mL/min. After that, the sample was loaded at the same flow rate and then 2 mL of H_2O were used to wash the cartridge sorbent, which was then left to dry under continuous air flow for 10 min and finally eluted with different organic solvents. After elution, the samples were evaporated to dryness under a gentle stream of nitrogen and reconstituted to 1 mL in the mobile phase (50:50 v/v MeOH:H_2O/MS). Solvent compatibility with MS/MS ESI and initial mobile phase composition was found to be critical for detection, reproducibility, signal intensity, and peak shape. The resulting solutions were filtered through a 0.45 μm nylon membrane syringe filters and analysed on the same day.

2.4. Method optimization

A One-Factor-at-a-Time (OFAT) [39] optimization methodology was employed to assess the critical parameters that affect the extraction, including: (i) the choice of SPE cartridges, (ii) the extraction time, (iii) the volume of solvent during extraction, (iv) the type of organic solvent used, and (v) the elution solvent. Additionally, the pH of the extract solution and the composition of the reconstitution solvent were evaluated, with a 50:50, v/v MeOH: H_2O mixture being the only composition that yielded quantifiable results. As BPs and PBs are lipophilic compounds with pK_a values ranging between 8–10 and 8–8.5 respectively, only non-polar solvents were tested while pH was evaluated over a broader range to evaluate whether it affects sample retention. For SPE, cartridge conditioning, sample loading, and elution were performed at a constant flow rate of 1 mL/min to achieve maximum yield. Each factor was tested in triplicate using blank-pad samples spiked with a mixture of 5 BPs and 4 PBs at 100 ng/mL, with relative recoveries used as the index of optimization. Fig. 1, illustrates the workflow of the proposed method.

2.5. Method validation

Linearity of the method was assessed over the range of 1–100 ng/mL ($n = 5$ levels, each analysed in triplicate) using ordinary least-squares regression. The coefficient of determination (R^2), slope, intercept, and associated errors are reported below. Sensitivity was evaluated using signal-to-noise (S/N) ratios, with the LOD and LOQ calculated as $LOD = 3 \times (S/N)$ and $LOQ = 10 \times (S/N)$, respectively. The signal was obtained from the peak area at the lowest validated concentration (1 ng/mL), while noise was measured within a defined baseline region adjacent to the analyte peak in a blank sample. Method accuracy and precision were evaluated at 1, 10, and 100 ng/mL for each analyte. Relative recovery (RR %) was calculated as the ratio of the mean concentration measured in samples spiked before extraction to the mean concentration measured in samples spiked after extraction, multiplied by 100 (Eq (1)). RSD% was calculated as the standard deviation divided by the mean measured concentration, multiplied by 100, and was used to assess intra-day repeatability ($n = 3$) and inter-day reproducibility (3 days × 3 replicates). Matrix effect (ME %) was evaluated at three concentration levels (10, 50, and 100 ng/mL). It was calculated as the ratio of the peak area obtained from blank extracts spiked after SPE to the peak area of neat standard solutions prepared at the same concentrations, multiplied by 100. Values below 100% indicated ion suppression, whereas values above 100% indicated ion enhancement. The mean ME (%) was calculated from triplicate measurements at each level (Eq (2)). ME was evaluated across different menstrual products to evaluate heterogeneity of the matrices.

Equation (1). Relative recovery % formula

$$\text{RR } (\%) = \frac{\text{amount found in pre} - \text{spiked sample}}{\text{amount found in post} - \text{extraction spiked sample}} \times 100$$

Equation (2). Matrix effect % formula

$$\text{ME } (\%) = \frac{\text{Peak area in post} - \text{extraction spiked sample}}{\text{Peak area in neat standard solution}} \times 100$$

2.6. LC-MS/MS conditions

The compounds were analysed and quantified using a Shimadzu (Tokyo, Japan) LCMS-8060RX system equipped with two LC-40D X3 binary pumps, a DGU-430 degasser unit, an SIL-40C X3 autosampler, a CTO-40C column oven, and an ESI source. An XBridge ® BEH C18 XP, (2.5 μm, 3.0 × 100 mm; Waters Corporation, Milford, MA, USA) column was utilized, thermostated at 40 °C, with a flow rate of 0.3 mL/min. A gradient elution program was applied using water (A) and methanol (B) as the mobile phase. The initial composition was set at 50:50 (v/v, A: B) and was linearly increased to 20:80 (v/v, A: B) over 10 min. An SPD-M40 photo diode array detector was also used, only for elution time and compound verification.

The mass spectrometric detection was performed using an electrospray ionization (ESI) interface operating in CoreSpray mode. The optimized interface parameters were as follows: nebulizing gas flow rate of 3 L/min, heating gas flow rate of 10 L/min, and drying gas flow rate of 10 L/min. The interface temperature was set at 300 °C, the desolvation line (DL) temperature at 300 °C, and the heat block temperature at 400 °C. The desolvation temperature was automatically set by the instrument's software at 526 °C. Multiple reaction monitoring (MRM) mode using negative ESI mode was performed where two transitions were selected for each analyte to ensure reliable identification and

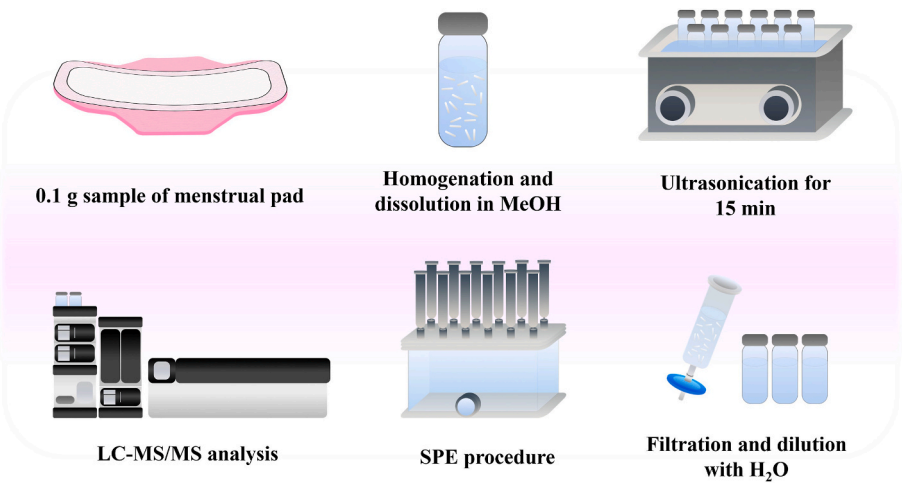


Fig. 1. Sample preparations steps of the proposed method.

quantification. MRM parameters were optimized individually for each compound by direct infusion of standard solutions while the most intense transition was used for quantification and a second transition was used for confirmation. For all analytes, dwell time was set at 50 ms with a pause time of 4 ms for quantifier and 1ms for qualifier ions. The final optimized precursor ions, product ions, retention times, and collision energies for all target compounds are summarized in Table S1. Fig. S1 shows the extracted chromatograms of each compound in a blank menstrual pad sample spiked at 1 ng/mL (LOQ).

DES was used as an internal standard at a constant concentration throughout the experiment, while the peaks of BPs and PBs were normalized accordingly. DES was added to the final extract after completion of the extraction and SPE clean-up steps and therefore functioned solely as an internal standard to correct for instrumental variability, injection repeatability, and ESI signal fluctuations. It should be noted that this approach does not correct for potential analyte losses and analyte-specific matrix effects during extraction and SPE purification. Therefore, DES served as an instrumental rather than a procedural internal standard, and extraction performance was assessed separately through method validation and recovery studies.

3. Results and discussion

3.1. Optimization results

As mentioned in Section 2, several parameters were optimized to achieve higher relative recoveries of analytes prior to method validation. With the extraction conditions set at 2 mL of solvent, 15 min of sonication, and 1 min of vortexing, several SPE cartridges were evaluated for their efficiency in adsorbing and retaining BPs and PBs during sample loading and subsequently eluting them with organic solvent. As shown in Fig. 2 (i), Bond Elut SAX, SCX, and FL cartridges showed minimal to no ability in retaining the targeted analytes and were therefore excluded from further investigation. Discovery DSC-18 and LICHROLUT RP-18 showed improved recovery rates, while HLB Oasis exhibited the highest recoveries (80-90%) and therefore it was selected for further optimization. This can be easily explained as C18 cartridges and HLB exhibit enhanced hydrophobic interactions with the aromatic

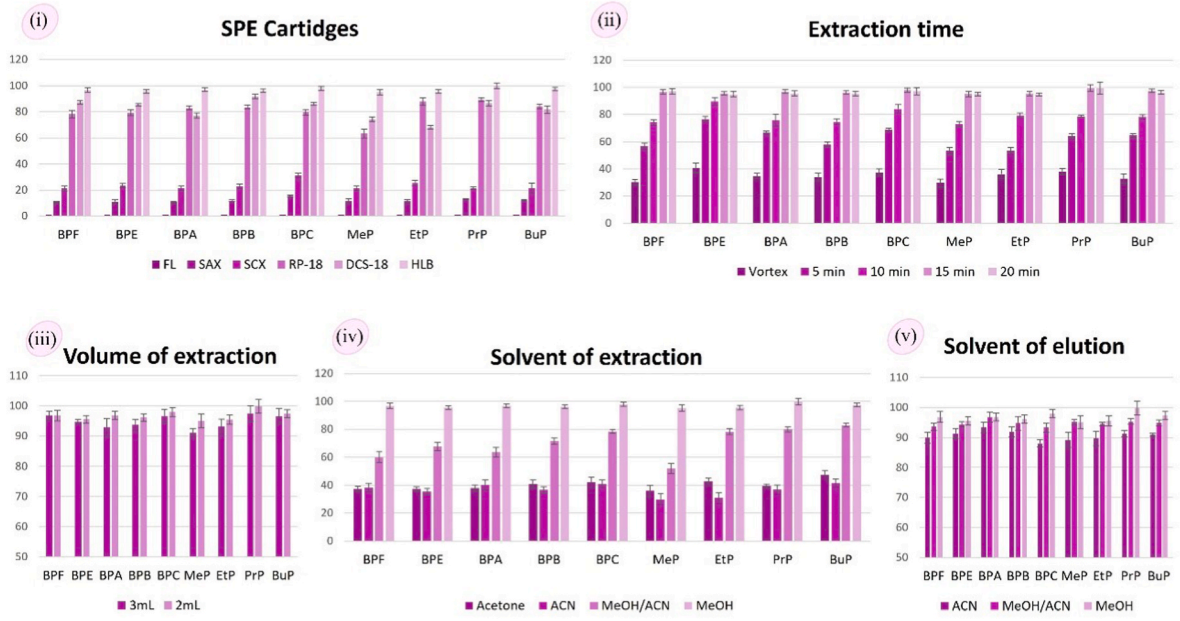


Fig. 2. Optimization results: (i) different SPE cartridges, (ii) extraction time of sample, (iii) volume of solvent used during extraction, (iv) solvent used during extraction, (v) solvent used during elution after SPE.

rings of BPs and PBs, thereby improving their retention. In addition, HLB (N-vinylpyrrolidone-divinylbenzene) provides broader polarity-range retention. Extraction time was evaluated by comparing simple vortexing for 1 min to sonication for 5, 10, 15, and 20 min Fig. 2 (ii). Vortexing alone resulted in minimal yields, below (<50%), whereas sonication caused a near-linear increase in recovery as sonication time increased. Sonication beyond 15 min showed little to no additional improvement, indicating that equilibrium had been reached. Therefore, an extraction time of 15 min was selected for subsequent experiments.

Solvent volume was tested on 0.1 g of sample, which was set the same across all experiments, where 1, 2, 3 and 5 mL of solvent were tested for extraction. Due to the adsorptive nature of menstrual pad materials, 1 mL of solvent was insufficient to fully wet the entire sample, while volumes above 3 mL were impractically large to handle. Furthermore, higher solvent volumes would require additional dilution prior to SPE, making the procedure less practical and less environmentally friendly and less aligned with green chemistry principles. Comparison of 2 mL and 3 mL solvent volumes showed no significant difference in recovery rates Fig. 2 (iii), leading to the selection of 2 mL, as the use of lower solvent volumes is preferable. To confirm complete extraction of the analytes, samples were re-diluted with an additional 2 mL of solvent, and the extraction procedure was repeated, demonstrating that the entire analyte content was effectively recovered.

As extraction solvents, MeOH, ACN, MeOH–ACN (1:1 v/v) and acetone were evaluated to identify the optimal organic solvent, with MeOH yielding the highest recoveries and acetone the lowest Fig. 2 (iv). BPs analysed have two phenol groups (–OH attached to benzene rings), while PBs have a single –OH group on the aromatic ring, making these compounds moderately polar. Among the solvents and solvent mixtures analysed, MeOH and acetone have relatively the lowest polarity. Methanol, being an alcohol, can act as both a hydrogen-bond donor and acceptor due to its –OH group [40], which makes it a protic solvent. This property allows MeOH to form extensive hydrogen bonds, which significantly enhances the solubility of BPs and PBs [41]. The presence of water should be avoided during any extraction step, and dilution prior to SPE should only be performed after collection and filtration of the extract, as hydrogels from the cellulosic fibres or polymers of the menstrual pad [20], can clog the SPE cartridges.

The pH value of the diluted methanolic extract was adjusted using formic acid and ammonium hydroxide and evaluated under acidic (pH = 3), neutral (pH = 7), and alkaline (pH = 9) conditions. As expected, alkaline solutions showed lower recoveries because BPs and PBs deprotonate at higher pH, become more soluble in water, and are therefore less retained on HLB Oasis cartridges. The presence of acid did not result in significant recovery enhancement and therefore, samples were kept at their original pH (~5–7) to simplify the analysis procedure.

After sample loading, SPE cartridges were washed with 2 mL H_2O–MS to remove unwanted interfering polar compounds and allowed to dry under vacuum for 4 min. Analytes were then eluted with an organic solvent. MeOH, ACN, and MeOH/ACN (1:1 v/v) were evaluated for elution efficiency, yielding almost similar recoveries Fig. 2 (v). ACN showed the lowest recovery, while MeOH and MeOH: ACN (1:1 v/v) gave higher, but comparable, recoveries. MeOH was selected, to minimized reagent versatility in the method. For all solvents, 2 × 1 mL aliquots were used for optimal analyte elution. To confirm that no residual analytes remained, the procedure was repeated, and no trace compounds (<LOD) were detected.

3.2. Validation results

For validation, the results were expressed in the extracts analysed in ng/mL. All analytes showed good linearity ($R^2 > 0.99$) over the concentration range of 1–100 ng/mL, as shown in Table S2. For all analytes, the LOQ was experimentally determined (S/N) to be ≤ 1 ng/mL, however, a value of 1 ng/mL was applied for consistency across validation and real samples. The LOD was determined to be 0.2 ng/mL and used in both validation and real sample analyses. At the 1 ng/mL level, the highest intra-day relative recovery (RR%) was observed for BPA (111.7%), while the lowest was for BPS (88.2%). Inter-day results at this level showed the highest recovery for MeP (107.3%) and the lowest again for BPS (84.1%). At 10 ng/mL, PrP showed the highest intra-day recovery (105.0%) and BPS the lowest (83.9%). For inter-day precision at 10 ng/mL, the highest recovery was obtained for BenP (106.3%) and the lowest for EtP (92.6%). At 100 ng/mL, BPF exhibited the highest recovery for both intra-day (112.2%) and inter-day (107.6%) measurements, whereas MeP showed the lowest recoveries at this level, with 83.9% intra-day and 85.5% inter-day. Recoveries for all analytes were within 80–120%, demonstrating good accuracy and method reliability, in accordance with the acceptance criteria recommended by method validation guidelines.

At the LOQ (1 ng/mL), all analytes showed RSD% below 20% for both intra-day and inter-day precision, with BuP exhibiting the highest value of 18.7%. Precision improved with increasing concentration, and at the mid (10 ng/mL) and high (100 ng/mL) validation levels, RSD% did not exceed 15%. Overall, the average RSD% across all three levels (1, 10, and 100 ng/mL) remained below 10%, demonstrating good method reproducibility. The validation data discussed above are presented in Table 1.

3.3. Matrix effect

The ME% for all analytes was generally within ±15% across the tested concentration levels (10, 50, and 100 ng/mL), as shown in Table S3. The highest ME% was observed for PrP at 109% (100 ng/mL), while the lowest was for BPF at 86.9% (10 ng/mL). Overall, the average ME% across all levels was approximately 99.5%, indicating minimal ion suppression or enhancement. These results indicate that the method is largely robust and not significantly affected by the sample matrix. Also, in Table S4, the average ME% (n = 3) measured at 50 ng/mL for different matrices is presented to evaluate variability in matrix effects among menstrual products (menstrual pads, panty liners, and tampons). The observed variability among all matrices was minimal, ranging from 88.5% to 113.2%, indicating that ME% did not significantly differ among products and therefore did not influence quantification under the optimized conditions.

3.4. Real sample analysis

The validated method was applied to real samples to demonstrate the practical applicability of the proposed method. Fortunately, as these compounds are considered a health risk, most analytes were either not detected or detected only at the LOD. Among the BPs, BPA was detected in all analysed samples, with concentrations ranging from the LOD up to 68.07 ng/g (in a tampon sample). In contrast, BPS, BPF, BPE, BPB, and BPC were rarely detected and mostly appeared at or below the LOD across samples (Table 2). Since BPA's presence in menstrual products would primarily result from contamination, leaching from resin/plastic components, or adhesive materials, it was expected to appear only at trace levels. BPA is a well-documented environmental contaminant due to its extensive production and widespread use in numerous consumer products [42,43]. Therefore, detection at or near the limit of detection (LOD) cannot conclusively confirm its presence and may only indicate minimal contamination. Samples were re-evaluated more than three times when bias was suspected, yet BPA consistently appeared. All other BPs, with the exception of BPS and BPE, showed no detectable signal and were therefore reported as not detected (n.d.). Procedural and solvent blanks were analysed throughout the analytical sequence to monitor potential laboratory contamination. Only BPA was detected in the procedural blank at levels below the LOQ of the proposed method, while no analytes were detected in solvent blanks prepared by repeating the procedure using solvents without any sample, confirming that laboratory contamination from reagents, solvents, or equipment did not

Table 1
Intra and inter day Relative Standard Deviation (RSD%) and Relative Recovery (RR%) of the analysed compounds at three different spiked concentrations (1, 10, 100 ng/mL).

Analyte	Spiked ng/mL	Intra day (n = 3)			Inter day (n = 3×3)		
		Found	RSD%	RR%	Found	RSD%	RR%
BPS	1	0.88 ± 0.06	6.82	88.22	0.84 ± 0.04	4.76	84.09
	10	8.39 ± 0.93	11.08	83.91	8.82 ± 0.89	10.09	88.19
	100	89.24 ± 3.08	3.45	89.24	93.85 ± 1.57	1.67	93.85
MeP	1	0.96 ± 0.11	11.46	95.81	1.07 ± 0.2	18.69	107.31
	10	8.74 ± 0.41	4.69	87.43	8.96 ± 0.49	5.47	89.58
	100	83.93 ± 6.7	7.98	83.93	85.48 ± 8.13	9.51	85.48
BPF	1	1.06 ± 0.13	12.26	105.99	0.93 ± 0.08	8.60	93.23
	10	9.96 ± 0.97	9.74	99.59	9.68 ± 1.21	12.5	96.85
	100	112.22 ± 3.5	7.98	112.22	107.56 ± 5.35	9.51	107.56
EtP	1	0.98 ± 0.14	14.29	98.19	1.03 ± 0.14	13.59	103.41
	10	9.37 ± 1.01	10.78	93.65	9.26 ± 0.59	6.37	92.61
	100	91.88 ± 8.53	9.28	91.88	94.01 ± 9.01	9.58	94.01
BPE	1	1.10 ± 0.14	12.73	110.18	0.99 ± 0.09	9.09	98.65
	10	8.54 ± 0.68	7.96	85.39	9.50 ± 1.13	11.89	94.97
	100	100.82 ± 2.72	2.70	100.82	104.8 ± 4.56	4.35	104.8
PrP	1	1.09 ± 0.11	10.09	109.33	1.06 ± 0.08	7.55	105.66
	10	10.5 ± 1.02	9.71	105.04	9.50 ± 0.88	9.26	94.99
	100	91.98 ± 5.76	6.26	91.98	97.03 ± 5.62	5.79	97.03
BPA	1	1.12 ± 0.03	2.68	111.68	0.99 ± 0.11	11.11	98.66
	10	8.87 ± 0.98	11.05	88.67	8.91 ± 0.58	6.51	89.11
	100	102.36 ± 5.91	5.91	102.36	100.03 ± 6.19	6.19	100.03
BenP	1	1.06 ± 0.02	1.89	105.96	1.05 ± 0.06	5.71	104.77
	10	10.44 ± 0.66	6.32	104.41	10.63 ± 0.16	1.51	106.27
	100	94.31 ± 4.05	4.29	94.31	99.11 ± 3.28	3.31	99.11
BPB	1	1.02 ± 0.02	1.96	102	1 ± 0.03	3.00	100.21
	10	9.06 ± 0.32	3.53	90.63	9.44 ± 1.29	13.67	94.35
	100	92.67 ± 7.42	8.01	92.67	96.18 ± 4.83	5.02	96.18
BuP	1	1.05 ± 0.16	15.24	104.54	1 ± 0.11	11.00	99.97
	10	10.48 ± 0.7	6.68	104.8	9.96 ± 0.32	3.21	99.58
	100	95.07 ± 3.27	3.44	95.07	100.09 ± 2.51	2.51	100.09
BPC	1	0.98 ± 0.1	10.2	98.1	0.99 ± 0.12	12.12	99.18
	10	9.20 ± 0.17	1.85	91.97	8.97 ± 0.53	5.91	89.73
	100	92.39 ± 10.88	11.78	92.39	94.15 ± 8.81	9.36	94.15

Table 2
Mean concentrations (ng analyte/g of sample) and standard deviations (±) of BPs determined in menstrual products.

Sample	BPS	BPF	BPE	BPA	BPB	BPC
Unscented m. pad	n.d.	n.d.	n.d.	<LOQ	n.d.	n.d.
Unscented m. pad	n.d.	n.d.	n.d.	<LOQ	n.d.	n.d.
Scented m. pad a	n.d.	n.d.	n.d.	15.55 ± 1.89	n.d.	<LOQ
Scented m. pad b	n.d.	n.d.	n.d.	<LOQ	n.d.	n.d.
Scented m. pad c	<LOQ	n.d.	n.d.	27.96 ± 1.88	n.d.	<LOQ
Scented m. pad d	<LOQ	n.d.	n.d.	24.95 ± 4.67	n.d.	<LOQ
Scented m. pad e	n.d.	n.d.	n.d.	44.97 ± 8.52	n.d.	<LOQ
Panty liner a	n.d.	n.d.	n.d.	11.12 ± 1.08	n.d.	n.d.
Panty liner b	<LOQ	n.d.	n.d.	25.72 ± 3.44	n.d.	n.d.
Panty liner c	<LOQ	n.d.	n.d.	22.91 ± 2.64	n.d.	n.d.
Tampon a	n.d.	n.d.	20.24 ± 0.74	17.05 ± 2.15	n.d.	<LOQ
Tampon b	<LOQ	n.d.	n.d.	17.01 ± 0.53	n.d.	n.d.
Tampon c	<LOQ	n.d.	n.d.	68.07 ± 5.42	n.d.	n.d.
Tampon d	<LOQ	n.d.	n.d.	22.32 ± 2.31	n.d.	<LOQ
Discontinued panty liner	<LOQ	<LOQ	n.d.	10.2 ± 2.13	n.d.	n.d.
Sterile gauge	n.d.	n.d.	n.d.	<LOQ	n.d.	n.d.
Toilet paper	n.d.	n.d.	n.d.	17.47 ± 0.98	n.d.	n.d.

*m. pad = menstrual pad **n.d. = not detected.

significantly affect the reported data.

In contrast to BPs, PBs are intentionally added as preservatives in a wide range of menstrual and personal care products [38,44] and were more frequently detected and at higher concentrations. Specifically, scented pads and panty liners showed elevated levels of MeP and PrP, with lower levels of EtP and BuP. BenP was not detected in any of the samples analysed. As shown in Table 3, the discontinued panty liner (sealed in its original package) exhibited 3–4 times higher concentrations of PBs, indicating that recent legislation has reduced the amounts of these compounds allowed in products. Sterile gauze, expected to contain no BPs or PBs, was used to assess method selectivity and to ensure that detections in commercial pads were not due to contamination or analytical bias, serving as an additional control blank.

3.5. Comparison with literature and green metrics

Previous studies on PBs and BPs in menstrual products reported comparable or higher concentrations than those found here. Gao and Kannan [17] analysed a variety of menstrual products, including menstrual pads, panty liners, and tampons, focusing on PBs, BPs, and related compounds. In their findings, PBs were frequently detected across all product types. MeP, EtP, PrP, and BuP ranged in concentration from

Table 3
Mean concentrations (ng analyte/g of sample) and standard deviations (±) of PBs determined in menstrual products.

Sample	MeP	EtP	PrP	BenP	BuP
Unscented m. pad	n.d.	n.d.	n.d.	n.d.	n.d.
Unscented m. pad	n.d.	n.d.	n.d.	n.d.	n.d.
Scented m. pad a	1461.69 ± 178.13	108.93 ± 2.97	770.01 ± 38.12	n.d.	104.1 ± 9.06
Scented m. pad b	n.d.	<LOQ	<LOQ	n.d.	n.d.
Scented m. pad c	<LOQ	<LOQ	17.82 ± 1.93	n.d.	n.d.
Scented m. pad d	73.69 ± 4.02	<LOQ	26.5 ± 3.04	n.d.	n.d.
Scented m. pad e	30.56 ± 4.81	<LOQ	4.72 ± 4.54	n.d.	n.d.
Panty liner a	n.d.	<LOQ	<LOQ	n.d.	n.d.
Panty liner b	<LOQ	<LOQ	32.05 ± 3.96	n.d.	<LOQ
Panty liner c	100.77 ± 17.53	21.6 ± 2.69	18.41 ± 1.08	n.d.	n.d.
Tampon a	<LOQ	n.d.	n.d.	n.d.	n.d.
Tampon b	42.62 ± 8.24	n.d.	<LOQ	n.d.	n.d.
Tampon c	n.d.	<LOQ	13.79 ± 2.55	n.d.	n.d.
Tampon d	17.83 ± 3.06	<LOQ	<LOQ	n.d.	n.d.
Discontinued panty liner	3635 ± 322.06	450.55 ± 91.24	2052.24 ± 24.39	n.d.	42.75 ± 8.48
Sterile gauge	n.d.	n.d.	n.d.	n.d.	n.d.
Toilet paper	n.d.	n.d.	n.d.	n.d.	n.d.

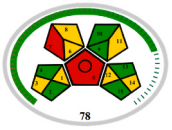
*m. pad = menstrual pad **n.d. = not detected.

LOD up to 245 ng/g in menstrual pads, up to 1240 ng/g in panty liners, and from LOD to 90 ng/g in tampons. BenP was detected only at concentrations close to the LOD. For BPs, BPA was prevalent in all samples, with concentrations ranging from <LOD to 55.9 ng/g in menstrual pads, <LOD to 157 ng/g in panty liners, and <LOD to 2.46 ng/g in tampons. Other BPs analysed (BPF and BPS) showed similar concentration ranges. In a more recent study, Blignaut et al. [37] reported substantial amounts of PBs, with 81% of panty liners (0.6–13.7 ng/g) and 75% of pads (0.6–13.7 ng/g) containing quantifiable levels, while BPs were detected in 75% of pantyliners (0.4–77.3 ng/g) and 100% of pads (0.4–77.3 ng/g). Similarly, studies analysing diapers, a closely related product to menstrual pads, reported comparable concentrations of PBs and BPs, with BPA concentrations reaching maximum levels of 200 and 520 ng/g (depending on the diaper component analysed). BPF and BPC were also detected at lower concentrations [45]. In another study concerning a variety of products, PBs in diapers were mainly detected as MeP and PrP (1.21–290 ng/g and up to 86.5 ng/g, respectively), while EtP, BuP, and BenP were quantified at lower concentrations [18].

In Table 4, a concise comparison of the proposed method's sample preparation, and analytical performance (RR%, RSD%, LOD) with the aforementioned methods is presented. All methods employed LC-MS/MS analysis. Compared to other studies that used higher amounts of organic solvents (5–40 mL of MeOH or less green solvents such as methyl tert-butyl ether) and extraction times exceeding in some 60 min with multiple steps, the proposed method is considered more environmentally friendly, rapid, and easily reproducible. As also shown, it can be easily adapted for similar personal products with satisfactory results.

Table 5 highlights the method's greenness, analytical performance, and practical applicability, as evaluated using the MoGAPI, RAPI, and BAGI tools. MoGAPI evaluates the environmental friendliness of an analytical method by considering multiple factors, such as sample pretreatment, storage and transport conditions, and the type and amount of chemicals used. A final score, with a maximum of 100, is calculated and classified as follows: Good (excellent green) ≥75%, Acceptable (moderate green) 50–74%, and Inadequate (poor green) < 50% [46]. The BAGI tool assesses practical applicability based on ten criteria, including type of analysis, number of analytes measured simultaneously, and instrumentation, producing both a numerical score (25–100) and a visual representation (a blue star), with scores above 60 indicating high applicability [47]. The Red Analytical Performance Index (RAPI) evaluates analytical performance across ten criteria (e.g., precision, accuracy, limit of detection), scoring each from 0 to 10 to generate a composite 0–100 score, which is visually represented as a red radial diagram [48].

Table 5
Analytical performance, green character and practicality of the proposed method based on RAPI, MoGAPI and BAGI assessment tools.

RAPI	MoGAPI	BAGI
		

Overall, the proposed method demonstrated a green character, with a MoGAPi score of 78, reflecting reduced solvent consumption and minimal sample preparation steps. Additionally, a BAGI score of 67.5, reflects good practical applicability of the method, supported by multi-analyte analysis, simultaneous sample preparation, the use of commercially available reagents and materials, and the requirement of only a small sample amount. The RAPI score of 77.5 was achieved due to the satisfactory analytical performance of the method, including good intermediate precision and reproducibility (RSD% = 1.5-18.6%), recovery (84-112%), matrix effect evaluation, good linearity, and selectivity of the method.

4. Conclusion

While most analysed samples showed minimal concentrations of BPs and PBs, frequent exposure could still increase the risk of adverse effects. Recent shifts toward organic-type products have led to labelling by

Table 4
Comparison of methods performance with literature.

Compounds	Matrix	Sample Preparation	RR%	RSD%	LOD (ng/mL)	Ref.
5BPs & 6 PBs	*M. products*	SLE	67-105	-	0.1–0.4	[17]
19 BPs	Diapers	UASE-PMSS	46-115	1.3-20	0.4-8	[45]
6 PBs	Paper pr.& diapers	SLE	76-108	-	0.2-0.5	[18]
7BPs & 5 PBs	Menstrual pads	SLE	66.2-93.8	3-9.5	0.008-1.7	[37]
6BPs & 5 PBs	*M. products*	SPE	84-112	1.5-18.6	0.2	Current

certification organization to ensure product safety. Transparency regarding product composition and maximum concentrations ensures that consumers are fully informed. Still, little research is available on the chemical composition and safety of these products, their everyday use, and the potential adverse effects they might cause.

This study proposed a rapid, green solvent-efficient, and sensitive (LOQ = 1 ng/mL) and practical method for the determination of endocrine disrupting chemicals found in menstrual pads. The method showed good performance with minimal RSD and high sensitivity (LOQ = 1 ng/mL), while also using readily accessible consumables such as SPE cartridges and can be reproduced easily. Evaluation with green metrics also supports these results, with scores above the threshold of greenness, practicality, and robustness. As for real sample analysis, a broad spectrum of menstrual products was evaluated and showed satisfactory results, while other samples, such as sterile gauze and toilet paper showed it can be easily adapted to similar adsorbent materials used in personal care products. As also shown, a drawback in the analysis of BPs and PBs is sometimes the ubiquitous presence of BPA, while for PBs, the easy contamination of samples from laboratory staff. Nonetheless, the proposed methodology may serve as a useful tool for future monitoring studies of endocrine-disrupting compounds in menstrual products. Further studies including a wider variety of personal care products and harmful compounds should be conducted to ensure consumer safety.

CRediT authorship contribution statement

Pavlos Voutsas: Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Natasa Kalogiouri:** Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – review & editing. **Victoria Samanidou:** Conceptualization, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aca.2026.345941.

Data availability

Data will be made available on request.

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